First Contact with Factorization Homology

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1 Introduction

Our aim through this section is to motivate a construction for a manifold invariant. For simplicity, we will focus on the 1-dimensional case in which we aim to construct an invariant for 1-dimensional manifolds from the geometry of embedding of 1-dimensional disks.

Definition 1.1. Let $\mathfrak{Disk}_{1,or}$ denote the category of oriented 1-dimensional disks which has objects consisting of finite disjoint union of oriented 1-dimensional open disks $(\varnothing, \mathbb{R}, \mathbb{R} \coprod \mathbb{R})$ and morphisms being all smooth embedding, orientation preserving maps, $\phi : \mathbb{R}^{\amalg k} \to \mathbb{R}^{\amalg m}$.

This category clearly comes with a natural symmetric monoidal structure with $\varnothing$ being the monoidal unit. Fix a field $\mathbb{K}$ and consider the symmetric monoidal category $\mathbf{Vect}_{\mathbb{K}}$ with tensor product $\otimes_{\mathbb{K}}$. Thus, we wish to study functors,

$$\mathcal{F} : \mathbf{Disks}^{\amalg}_{1,or} \to \mathbf{Vect}^{\otimes_{\mathbb{K}}}_{\mathbb{K}}$$

such that $\mathcal{F}$ is symmetric monoidal and we further wish for isotopic embeddings to be sent to the same linear map. Thus, we introduce a notion of homotopy of maps that we wish our functor to preserve.

Definition 1.2. Two embeddings $f_0, f_1 : \mathcal{M} \to \mathcal{N}$ are isotopic if there exist a continuous family of embeddings $f(t) : \mathcal{M} \to \mathcal{N}$, $t \in [0,1]$ s.t $f_0 = f(0)$ and $f_1 = f(1)$.

This means that we can homotopically deform f_0 to f_1 by a series of embeddings. For instance, one can consider the orientation preserving smooth embeddings $\mathbb{R} \to \mathbb{R}$, this is the space of strictly increasing functions which is isotopic to the identity. Since we want to construct a functor that is symmetric monoidal, this fixes its action on disjoint union,

$$\mathcal{F}(\bigsqcup_k \mathbb{R}) = \bigotimes_k \mathcal{F}(\mathbb{R})$$

Theorem 1.1. *[1] Suppose $\mathcal{F} : \mathfrak{D}isk^{\sqcup}_{1,or} \to \mathbf{Vect}^{\otimes_{\mathbb{K}}}_{\mathbb{K}}$ is a symmetric monoidal functor such that if ϕ_1 and ϕ_2 are isotopic embeddings then, $\mathcal{F}(\phi_1) = \mathcal{F}(\phi_2)$. Then the data of $\mathcal{F}$ is equivalent to the data of a unital associative $\mathbb{K}$-algebra*

Proof. (sketch) Since $\mathcal{F}$ is symmetric monoidal it forces $\varnothing \mapsto \mathbb{K}$. Let $A = \mathcal{F}(\mathbb{R})$, then,

$$\mathcal{F}(\bigsqcup_k \mathbb{R}) = A^{\otimes k}$$

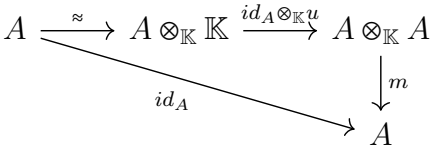
Now we need to check its action on morphisms. Consider the morphism set $\mathbf{Hom}_{\mathfrak{Disk}_{1,or}}(\varnothing, \mathbb{R})$. This clearly has only one element $\eta : \varnothing \to \mathbb{R}$. Any other embedding would be isotopic to this map. By monoidality of $\mathcal{F}$ this map would be sent to the unit map $u = \mathcal{F}\eta : \mathbb{K} \to A$. Now to prove that this is an associative algebra we need a candidate for the multiplication map. Thus, we need to consider the set $\mathbf{Hom}_{\mathfrak{Disk}_{1,or}}(\mathbb{R} \coprod \mathbb{R}, \mathbb{R})$. Up to isotropy, there are exactly two such non isotopic embeddings which send copy of $\mathbb{R}$ in front of the other.

So we have two maps ϕ_1, ϕ_2. Let $\phi_1 : \mathbb{R} \bigsqcup \mathbb{R} \to \mathbb{R}$ be the orientation preserving embedding that preserve the order $\mathbb{R}_1 \to \mathbb{R}_+$ and $\mathbb{R}_2 \to \mathbb{R}_-$. We choose this as our multiplication map. So let,

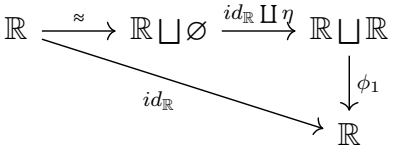
$$m = \mathcal{F}(\phi_1) : A \otimes_{\mathbb{K}} A \to A$$

Now, we need to show that this multiplication map determines an unital associative algebra structure on A.

- Unit: consider the following diagram,

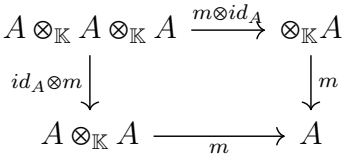


which corresponds in $\mathfrak{Disk}_{1,or}$ to

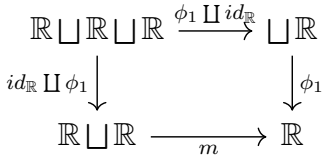


This map does not need to be an identity, however, it is up to isotropy.

- Associativity: In a similar manner, we need to check the commutativity of the following diagram,



similarly, due to isotropy requirement, the corresponding diagram,



commutes. Thus, we have built a unital associative algebra. Now for the other embedding map ϕ_2, One can see that $\phi_2 = \phi_1 \circ \sigma$ where σ is the symmetric braiding. Thus, it adds no new data. Using the same construction as above a unital associative algebra can give rise to a functor $\mathcal{F}$. Thus, up to isotopy all embeddings can be factored as a composition of 1) Component permutations, 2) disjoint union of η and ϕ_1. Furthermore, this is an equivalence of categories, thus a natural transformation κ between functors $\mathcal{F}$ and $\mathcal{G}$ gives rise to a unital algebra morphism $A \to B$. This directly follows from the naturalality of κ.

■

Guided by this intuition, we will call every such functor from Disks with this isotropy property an algebra of disks. We will make this definition precise later on. Given this disk algebra, we wish to construct an invariant of all 1-dimensional manifolds.

Definition 1.3. Let $\mathfrak{Mfld}^{\sqcup}_{1,or}$ be the category of oriented 1-dimensional manifolds with finitely many connected components and morphisms are smooth embeddings that respect orientation. Its again symmetric and monoidal and we have $\mathfrak{Disk}_{1,or} \subset \mathfrak{Mfld}_{1,or}$.

Thus, we are searching for an extension of $\mathcal{F}$,

$$\begin{array}{ccc} \mathfrak{Disk}_{1,or} & \xrightarrow{\mathcal{F}} & \mathbf{Vect}_{\mathbb{K}} \\ \downarrow & \nearrow & \\ \mathfrak{Mfld}_{1,or} & & \end{array}$$

For now, let us suppose that such an extension indeed exists. Furthermore, let us denote $\mathcal{F}(S^1)$ the vector space of S^1 assigned by the extension. The only data given to us is that of intervals. By the fact that S^1 is a gluing of an interval, intuitively, we expect a similar kind of gluing of the corresponding vector space of the disk. In general, we can have embeddings $\mathbb{R}^{\sqcup k} \to S^1$. Then we must have a map $\phi_k : A^{\otimes k} \to \mathcal{F}(S^1)$. However, again all these embeddings must all factor through A, i.e

$$\phi_k : A^{\otimes k} \to A \to \mathcal{F}(S^1)$$

One can further notice that there is only one class of embeddings $\phi_2 : \mathbb{R} \sqcup \mathbb{R} \to \mathcal{F}(S^1)$. Thus, one can find an isotopy between, $\phi_2 \circ \sigma$ and ϕ_2 by moving around the circle. Thus, let $x_1, x_2 \in A$ and consider,

$$\phi_1(x_2 x_1) = \phi_2(x_2 \otimes x_1) = \phi_2(x_1 \otimes x_2) = \phi_1(x_1 x_2)$$

This just tells us that the vector space that should be associated with S^1 is endowed with a commutative algebra. If $[A, A] \subset A$ is the vector subspace generated by the commutators $x_1 x_2 - x_2 x_1$, then

$$\mathcal{F}(S^1) = A/[A, A]$$

In that sense, we have a global invariant of 1-dimensional manifolds from local data associated with the disks.

This definition of a global invariant can be extended to higher dimensions. This is what is known as factorization homology. Factorization homology is defined as a left Kan extension, following the established definition in [2]. Factorization homology takes as an input an n-dimensional manifold with an E_n algebra which is an algebra over the little disc operad over a symmetric monoidal ∞-category $\mathcal{C}$. Ayala and Francis proved that factorization homology satisfies Eilenberg-Steenrod axioms for singular homology [2]. Thus, one may formally consider factorization homology as a generalization of singular homology made for topological manifolds. In particular, the factorization homology satisfies excision. Furthermore, it was

shown that depending on the algebraic input one can obtain different homology theories [3]. In that sense, factorization homology associates a manifold invariant by associating to a disc system of local coordinates in an ∞-category and integrates this over the manifold (Meaning we are extending the data of disk algebra functor to the whole manifold).The factorization homology of $\mathcal{M}$ with coefficients in $\mathcal{A}$ is an object in $\mathcal{C}$ given by the left kan-extension as,

$$\int_{\mathcal{M}} \mathcal{A} = colim(\mathfrak{D}\mathrm{isk}_{n/\mathcal{M}} \to \mathfrak{D}\mathrm{isk}_n \to \mathcal{C})$$

where, $\mathfrak{D}\mathrm{isk}_{n/\mathcal{M}}$ is essentially the category over $\mathcal{M}$ considered as simplicial set.

Factorization homology satisfies two main important properties. For a collar gluing $\mathcal{M}' \bigcup_{M_0 \times \mathbb{R}} \mathcal{M}''$ we have that each factor $\int_{\mathcal{M}'} \mathcal{A}$ and $\int_{\mathcal{M}''} \mathcal{A}$ is a module over the algebra $\int_{M_0 \times \mathbb{R}} \mathcal{A}$. Additionally, there is an equivalence in $\mathcal{C}$,

$$\int_{\mathcal{M}'} \mathcal{A} \bigotimes_{\int_{M_0 \times \mathbb{R}} \mathcal{A}} \int_{\mathcal{M}''} \mathcal{A} \to \int_{\mathcal{M}} \mathcal{A}$$

Furthermore, factorization is a pointed theory meaning it comes with a canonical E_0-structure. This can be constructed as follows, Let $\varnothing$ be the empty manifold, since $\varnothing$ is the monoidal unit in $\mathfrak{M}\mathrm{fld}_n$ and the factorization is a monoidal functor, then have the isomorphism,

$$\int_{\varnothing} \mathcal{A} \approx 1_{\mathcal{C}}$$

But we notes that $\varnothing$ is initial in $\mathfrak{M}\mathrm{fld}_n$. This directly translates into a unique morphism,

$$1_{\mathcal{C}} \to \int_{\mathcal{M}} \mathcal{A}$$

by the factorization homology. Although this embedding may seem trivial (it gives no extra information) it essentially completely determined the factorization homology for surfaces.

Factorization homology for surface takes an algebraic input which is essentiality a 2-disk algebra valued in a symmetric ∞-category $\mathcal{C}^{\boxtimes}$ and a surface Σ. There are in general different versions of disk algebras depending on the tangential structure associated with the disks. We focus on framed E_2 algebras which are the algebras associated with oriented disks and orientation-preserving embeddings. One can formally construct such a category by noticing that a punctured surface can be constructed by gluing handles on a marked disk. Thus, technically speaking we split the surface into three parts, as a set of g-handles, crossings, and a disk. Then using the excision property, one can formally calculate the factorization homology of the surface. One then has the following result for $\mathcal{A}$ an abelian rigid balanced braided tensor category in **Rex**. We have an equivalence of categories,

$$\int_{\Sigma(P)} \mathcal{A} \cong A_{\Sigma(P)} - mod_{\mathcal{A}}$$

where $\Sigma(P)$ is the surface constructed using gluing pattern P and $A_{\Sigma(P)} = End(\mathcal{O}_{\mathcal{A},\Sigma(P)})$.

In this thesis we will give a brief discussion of the above notions. We will introduce a model for infinity small categories. Then, discuss the framed little operad. Lastly, we will point out the properties of factorization homology.

2 Categorical Constructions

In this section, we will a brief discussion for the constructions need for the definition of factorization homology. Thus, we will construct a model to encode homotopy between maps in category theory. We will mainly be following [4] [5]. For details about the given introduction one can see [6]. Our main objective is to be able to identify objects which are not isomorphic in a purely categorical sense but are equivalent from a homotopy theoretic point of view. Thus, we wish to think of these homotopy equivalences as actual isomorphisms. Thus, we have to accept that in some sense we need to add additional information not given by the underlying categorical structure. This is indeed a valid way which lead to the Grothendick's theory of derivators [7]. However, this approach is not very satisfying since we wish to hold on the idea that the homotopy version of our category should be manifested through some universality property.

Nonetheless, homotopies just seem like morphisms. So basically in our category we have objects, morphism and morphisms between morphisms (homotopies). But there is a huge problem in this approach; homotopies are not unital nor associative. So, how could we solve such a problem? One can for instance consider homotopy classes instead of homotopies and indeed we will end up with a 2-category. However, again this approach does not provide a universal justification. Thus, it seems that we need to recite all the information of morphisms, meaning we need also to consider homotopy between homotopies, homotopy between homotopies of homotopies and so on. This is an informal definition of what is an $(\infty, 1)$-category is. Through this thesis, we will be using ∞-category to denote $(\infty, 1)$-category. Here, the infinity symbol refers to this tower of morphism while the 1 refers to the fact all higher morphism of homotopies are invertible.

There are a lot of models that were constructed to describe such a definition, including quasi-categories, model categories, Complete Segal spaces, simplicial categories and many others [8] [6]. However, we will be giving a brief non formal introduction to the simplicial sets which best serves our purposes.

2.1 Simplicial Sets

In this section, we give a brief introduction to simplicial sets. Simplicial sets are essentially generalizations of the geometric simplicial complexes of elementary algebraic topology. One can look at [9] which adapts this approach. Here we will be giving the basic definitions,

Definition 2.1. A simplicial set consists of a sequence of sets $X_0, X_1, \cdots$ and for each $n \geq 0$, functions $d_i : X_n \to X_{n-1}$ and $s_i : X_n \to X_{n+1}$ for each $0 \leq i \leq n$. These maps are called face and degeneracy maps respectively. The maps satisfy the following relations,

$$\begin{aligned}
d_i d_j &= d_{j-1} d_i, && i < j \\
d_i s_j &= s_{j+1} d_i, && i < j \\
d_j s_j &= d_{j+1} s_j = id && \\
d_i s_j &= s_j d_{i-1}, && i > j+1 \\
s_i s_j &= s_{j+1} s_i && i \leq j
\end{aligned}$$

One can intuitively think of the set of a simplicial set as a collection of n-simplices with the usual face and degeneracy map. In that case, the simplicial set is describing the whole complex.

Example 2.1. Any simplicial complex can be made into a simplicial set by adding all possible degeneracy simplices. For example, the standard $X = [0,1]$ simplex has $n-2$ elements in each X_n. For example, $X_2 = \{[0,0,1],[0,1,1],[0,0,0],[1,1,1]\}$.

Remark. We will refer to the standard ordered n-simplex thought of as a simplicial set by $\Delta^n = [0,1,\cdots,n]$

Example 2.2. Let X be a topological space. Let $|\Delta^n|$ denote the standard n-simplex and $\mathcal{C}(X)_n$ the set of continuous functions $|\Delta^n| \to X$. This makes a simplicial set called the singular set of X. One can further use these sets to construct the singular chain complex $\mathbf{Sing}|X|$ of topological space X that has $\mathbf{Sing}|X|_n$ equal to the free abelian group generated by $\mathcal{C}(X)_n$. To see that this is indeed a simplicial set one can look at the action of the face and degeneracy maps on a continuous function $\sigma : |\Delta^n| \to X$. The $d_i\sigma$ is the restriction of σ to the ith face of $|\Delta^n|$. Equivalently, one can describe this as the composition of σ and the simplicial inclusion map $[0,\cdots,n-1] \to [0,\cdots,\hat{i},\cdots,n]$.

On the other hand, s_i take the singular simplex σ to the composition of σ with collapse represented by degeneracy $[0,\cdots,n-1] \to [0,\cdots,i,i,\cdots,m]$

Definition 2.2. A simplex $x \in X_n$ is called non degenerate if x cannot be written as $s_i y$ for $y \in X_{n-1} \forall i$.

now we give the definition of simplicial sets from a categorical point of view.

Definition 2.3. The category Δ has objects as finite ordinal sets $[n] = \{0,1,\cdots,n\}$ and morphisms of Δ are order preserving functions $[m] \to [n]$.

Basically, the objects of Δ are prototypes for the n-simplices set. An example of a morphism $f : [5] \to [3]$ can be described by $f[0,1,2,3,4,5] = [0,0,2,2,3,3]$ which takes a 5-simplex degenerately into a 3-simplex. One can then see that all morphism in Δ are just some compositions of $D_i[0,\cdots,n] = [o,\cdots,\hat{i},\cdots,n+1]$ inclusions and $S_i[0,\cdots,n+1] = [0,\cdots,i,i,\cdots,n+1]$ collapsing maps.

If we wish to use the ordinals to describe the whole set X_n we need a way to describe the face and degeneracy maps. Its clear from the geometrical intuition that the face and degeneracy maps are opposite to inclusion and collapsing maps respectively. Thus, we must consider Δ^{op}. One can then easily reach the following definition,

Definition 2.4. A simplicial Set is a contravariant functor functor $X : \Delta^{op} \to \mathbf{Set}$

Example 2.3. The singular set $\mathcal{C}(X)$ is a functor $\Delta^{op} \to \mathbf{Set}$ that assign to $[n]$ the set $\mathbf{Hom_{Top}}(|\Delta^n|, X)$. Furthermore, $\mathcal{C}(X)(d_i) : \mathbf{Hom_{Top}}(|\Delta^n|, X) \to \mathbf{Hom_{Top}}(|\Delta^{n-1}|, X)$. Similarly for s_i.

Thus basically, one can form a functor category of simplicial sets where morphisms are just natural transformations of functors.

Remark. A simplicial morphism need only to be defined on non degenerate simplices. This is mainly due to the naturalality conditions, $f(s_i(x)) = s_i(f(x))$ similarly for face maps.

2.2 Kan complexes

One of the goals of the development of simplicial sets and other simplicial objects was to find a combinatorial way to study homotopy theory [10]. This is the same idea that one sees in simplicial homology where it allows us to derive invariants of simplicial complexes in a combinatorial manner. However, it has been shown that we need to impose extra conditions on simplicial sets to make them appropriate for the study of homotopy theory.

Definition 2.5. The k-th horn on an n-simplex $|\Delta^n|$ is the subcomplex of $|\Delta^n|$ obtained by removing the interior of $|\Delta^n|$ and the interior of the face $d_k\Delta^n$.

Let Λ^n_k refer to the corresponding simplicial set of the k-th horn. Now, the condition we need to impose on the simplicial sets is known as the kan condition which says that whenever we see a horn on an n-simplex within a simplicial set the whole simplex also exists in the simplicial set.

Definition 2.6. The simplicial object X satisfies the extension condition or kan condition if for any morphism of simplicial sets $\Lambda^n_k \to X$ can be extended to a simplicial morphism $\Delta^n \to X$

$$\begin{array}{ccc} \Lambda^n_k & \longrightarrow & X \\ \downarrow & \nearrow & \\ \Delta^n & & \end{array}$$

Example 2.4. Δ^n the standard simplices does not satisfy the kan condition. Consider for example $\Delta^1 = [0,1]$ and the horn Λ^2_0 which has edges $[0,2]$ and $[0,1]$ of Δ^2 with their degeneracies. Since its sufficient to consider the action of the map on non degenerate simplices, one can see that the map $\Lambda^2_0 \to \Delta^1$ given by,

$$[0,2] \mapsto [0,0]$$

$$[0,1] \mapsto [0,1]$$

This map is well defined since as a simplicial map it is order preserving (we do not have to compare 1 and 2 since that edge is missing). But, this map cannot be extended to the whole 2-simplex since we have already identified $0 \mapsto 0$, $2 \mapsto 0$ and $1 \mapsto 1$ which is not order preserving in Δ^2.

Example 2.5. Given a topological space X, the simplicial set $\mathcal{C}(X)$ does satisfy the kan condition. Consider $f : \Lambda^n_k \to \mathcal{C}(X)$. This map is the same a specifying for each $n-1$ faces $d_i\Delta^n$ of Δ^n a singular complex $\sigma_i : \Delta^{n-1} \to X$ such that $i \neq k$. Since every other simplex of Λ^n_k is a face or a degeneracy of a face, the whole map is completely determined by the data of these maps. From the simplicial set axioms, these σ_i can be made to glue together consistently using the face and degeneracy map to give rise to a continuous map $f : |\Lambda^n_k| \to X$. Now using this map we want to extend it to a map for $|\Delta^n|$. Let $\rho : |\Delta^n| \to |\Lambda^n_k|$ be any continuous retraction. Then, the required map is $f \circ \rho : |\Delta^n| \to X$.

2.3 ∞-Categories

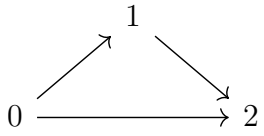
Now, we are ready to motivate a definition for ∞-categories. Recall, that we wanted our category to have this notion of homotopy with it. However, this also means that if we can give a model for ∞-categories, ordinary categories should also fit into that definition by just putting all higher morphisms to identities. Thus, to be able to realise an adequate definition, we need to look at ordinary categories from a different point of view, specifically from the point of view of a simplicial set.

Definition 2.7. Let $\mathcal{C}$ be a small category. The nerve of $\mathcal{C}$ is the simplicial set defined by,

$$N(\mathcal{C})_n = \mathbf{Hom}_{\mathbf{Cat}}([n], \mathcal{C})$$

where $[n]$ is the ordinal considered as a category. Basically, $N(\mathcal{C})_n$ is the set of strings of n-composable morphisms in $\mathcal{C}$.

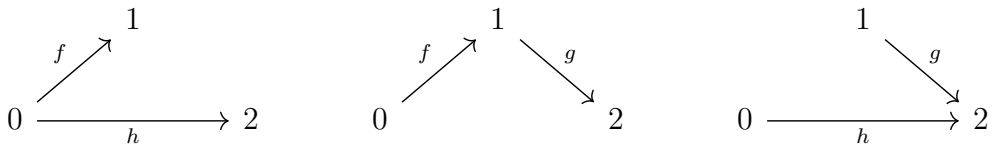
Consider for example $N(\mathcal{C})_2$, this is just the set of functors defined on the digram,



If one considers the face map $d_1 : N(\mathcal{C})_2 \to N(\mathcal{C})_1$, this would just give us the leg defined on 0–2. But, this is just the composition of the two morphisms. Thus, one can see that a 2-simplex in $N(\mathcal{C})$ is just a pair of morphism and their composition.

Lemma 2.1. *([11],1.3.3.1) The nerve functor* $\mathbf{Hom}_{\mathbf{Cat}}([n], -) : \mathbf{Cat} \to \mathbf{sSet}$ *is fully faithful and hence induces an equivalence onto its essential image.*

This just mean that all of **Cat** is just a subcategory of **sSet**. Thus, we wish to describe this essential image of the nerve. Particularly, we want to know when is a simplicial set S isomorphic to the nerve of some category. So, we need to see interaction of S with other simplicial sets. One of particular interest is the boundary of the simplicial set or the k-th horn defined above. Let us illustrate this property in dimension 2. Consider the simplicial map $\lambda : \Lambda^2_k \to N(C)$. Recall that a simplicial map is natural transformation between these two functors. In particular one recalls in the inclusion of the k-th horn inside the full simplicial set. Thus, in $\mathcal{C}$, one gets the following digrams,



However, if one looks closely at the middle digram. This diagram can be filled in $\mathcal{C}$ by the composition $g \circ f$. This means that any small category has a horn filling property of

this sort. For the other two horns Λ^2_0, Λ^2_2, it seems that they require the existence of inverse images of g and f ($h = id$) which may not be found in all categories. Thus, one wonders if the composition associated to the category necessitates the this filling property for higher dimensions for $0 < k < n$ and whether this is enough to characterise a nerve of some category. This is answered in the following proposition.

Proposition 2.1. ([11], 1.3.4.1) Let S be a simplicial set. Then S is isomorphic to the nerve of a category if and only if for every pair of integers and every map of simplicial sets $\sigma_0 : \Lambda^n_i \to S$ for $0 < i < n$, there exists a unique map $\sigma : \Delta^n \to S$ such that $\sigma_0 = \sigma|_{\Lambda^n_i}$.

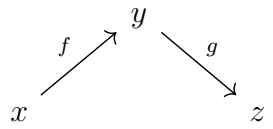
This just reminds us of the Kan complex. However, notice the main difference here. The Kan extension property was a mere existence theorem (it does not need to be unique). Furthermore, all horns can be extended in a Kan complex. So, this motivates the following,

Definition 2.8. The horns Λ^n_k for $0 < k < n$ are called inner horn while Λ^n_0 and Λ^n_n are called outer horns.

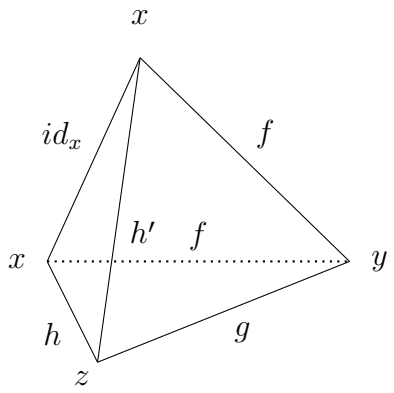
Now, we ready to introduce the homotopy notion in our category. So, again, we want to define this deforming behaviour of morphisms. In particular, a composition of g and f can be homotopic to a whole class of morphisms. Thus, one way to introduce this notion is basically to get rid of this uniqueness property.

Definition 2.9. A simplicial set $\mathcal{C}$ is an ∞-category if every inner horn $\Lambda^n_k \to \mathcal{C}$, $0 < k < n$ can be extended to an n-simplex $\Delta^n \to \mathcal{C}$.

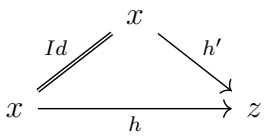
Let us now see how this definition matches our requirements. One can think of an ∞-category in the following way. Objects are vertices $x \in \mathbf{C}_0$, morphisms are 1-simplices $f \in \mathcal{C}_1$. In the sense the face map $d_1 : \mathcal{C}_1 \to \mathcal{C}_0$ is just the source map and $d_0 : \mathcal{C}_1 \to \mathcal{C}_0$ is the target map. Now consider $x \in \mathcal{C}_0$, we want an identity map. Recall that $d_0 s_0 = d_1 s_0 = id_{\mathcal{C}_0}$. Thus, naturally, one can consider the degeneracy map $s_0 : \mathcal{C}_0 \to \mathcal{C}_1$ as an identity of every object $x \in \mathcal{C}$. As for compositions, one can consider two morphisms $f : x \to y$ and $g : y \to z$. Graphically, they define the following horn,



This is an inner horn which from the definition can be extended to a full 2-simplex $\sigma : \Delta^2 \to \mathcal{C}$. This in some sense gives us the composition. However, one has to note that there is no composition in an ∞-category since this map is not unique. Instead one demands that there is a way to compose arrows. Furthermore, one typically expects that the space of all such maps that can fill in the digram is contractible. So how can the construction give us a homotopy? Consider two different simplices $\sigma, \sigma' : \Delta^2 \to \mathcal{C}$ such that they have candidate composition maps with $h \neq h'$. Let $S \subset \{0, \cdots, n\}$, we will use the notation that $\Delta^S \subset \Delta^n$ to denote the $|S|$-dimensional face of Δ^n whose vertices are S. Consider the degenerate triangle τ defined by $\tau|_{\Delta^{1,2}} = \tau_{\Delta^{0,2}} = f$ and $\tau_{\Delta^{0,1}} = s_0 x$ which is degenerate on x (we emphasize here the we can canonically define a map form $s_0 x$ to y since it's defined on non degenerate simplex). We can glue the maps τ, σ and σ' into a Λ^3_2 horn shown as follows,

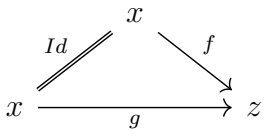


In particular again due to the horn extension property, we can extend this to fill in the last face. Thus, one gets,



We will take this as our definition of homotopy between morphisms.

Definition 2.10. Let $f, g : x \to y$ be two morphisms in $\mathcal{C}$ an ∞-category. f and g are homotopic if there exist a 2-simplex $\sigma : \Delta^2 \to \mathcal{C}$ with,



In that sense, composition is not strictly defined, it is defined up to homotopy. Furthermore it has been shown that the collection of the compositions defines a contractible Kan complex [6]. One also needs to check that defines an equivalence relation,

Proposition 2.2. ([4], 1.13) Let $\mathbf{C}$ be an ∞-category and let $x, y \in \mathcal{C}_0$. The homotopy relation is an equivalence relation on $\mathbf{Hom}_{\mathcal{C}}(x, y)$ and the homotopy class of morphisms $f : x \to y$ is denoted $[f]$.

Now that we have defined what homotopies are, we still need notions of homotopies between homotopies and so on. One can just generalize the above notion; An n-morphisms from x to y is a map of simplicial sets,

$$\tau : \Delta^{n+1} \to \mathcal{C}$$

such that $\tau|_{\Delta^{0,\cdots,n}} = x$ (through as a degenerate simplex $s_o^n x$) and $\tau|_{\Delta^{n+1}} = y$.

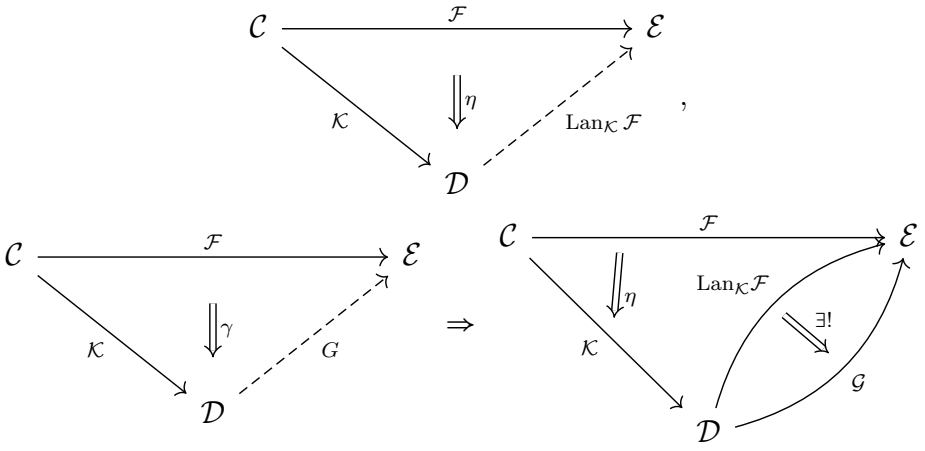
Remark. One last requirement we need to check is that homotopies have inverses.

Now that we have an idea of what is an infinity small category. One should introduce the notion of the monoidal product inside an ∞-category. A symmetric monoidal category is just a category with a bi-functor that needs to satisfy the unit and hexagon axioms. To upgrade this, on typically needs change these strict equations into their homotopy version. However, this is not a direct task since we would a lot of coherence equation. These coherence equations are organised by the notion of fibrations. We will use this initiative idea to work with symmetric monoidal infinity categories. An interested reader should check [5] [6].

2.4 Kan Extensions

In this section, we will be describing the way to extend functors given the data associated to other functors. The data of a kan extension consists of a functor and a natural transformation. Kan extensions as the name suggests is a way to expand the behaviour of a functor to another category filling in the missing gaps. We will review classical notions of Kan extensions then one can generalize this notion to the homotopy version by extending in the simplicial set version of the category.

Definition 2.11. Given functors $\mathcal{F} : \mathcal{C} \to \mathcal{E}$ and $\mathcal{K} : \mathcal{C} \to \mathcal{D}$, a left kan extension of $\mathcal{F}$ along $\mathcal{K}$ is a functor $Lan_{\mathcal{K}}\mathcal{F} : \mathcal{D} \to \mathcal{E}$ together with a natural transformation $\eta : \mathcal{F} \implies Lan_{\mathcal{K}}\mathcal{F} \circ \mathcal{K}$ such that for any other pair $(\mathcal{G} : \mathcal{D} \to \mathcal{E}, \gamma : \mathcal{F} \implies \mathcal{G} \circ \mathcal{K})$ γ factors uniquely through η.



Remark. Dually, there is notion of right kan extensions. However, we are only interested in the left kan extensions.

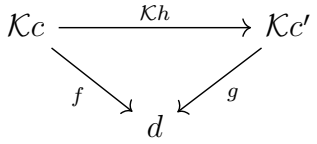
This definition captures the properties we want our extension to have that it is the best candidate of all possible functors. Now, we want a way to define the functor $Lan_{\mathcal{K}}\mathcal{F} : \mathcal{D} \to \mathcal{E}$. We wish to define the value of $Lan_{\mathcal{K}}\mathcal{F}(d)$ where $d \in \mathcal{D}$. In some sense, one need to consider all the possible approximation of the element d that comes from the category $\mathcal{C}$. After which, one can apply functor $\mathcal{F}$. One can see that the ideal case is when $\mathcal{F} \cong Lan_{\mathcal{K}}\mathcal{F} \circ \mathcal{K}$ but this is not usually possible since one mostly have the situation where $\mathcal{K}$ embeds $\mathcal{C}$ in $\mathcal{D}$ in some manner. Thus, typically, $\mathcal{D}$ has objects and morphisms which are not in the image of $\mathcal{K}$. However, we need somehow to define the action of $Lan_{\mathcal{K}}\mathcal{F}$ on these objects.

To solve this problem one can recall the Yoneda embedding. We know that all the information about an object is encoded in totality of morphisms from or to the object (i.e

Hom$(-, d)$ and **Hom**$(d, -)$). So, one way to approximate the object d is not to consider all such morphisms but rather only those which are images of $\mathcal{K}$. If we can do so we can back propagate this information to $\mathcal{C}$ and we are done. Let us define a category based on this information.

Definition 2.12. Given a functor $\mathcal{K} : \mathcal{C} \to \mathcal{D}$ and $d \in \mathcal{D}$, the comma category $\mathcal{K} \downarrow d$ has,

- Objects: (c, f) where $c \in \mathcal{C}$ and $f : \mathcal{K}c \to d$.
- Morphisms: $h : (c, f) \to (c', g)$ are such that,



Now we wish to back propagate the information. One can define the projection functor $\Pi_d : \mathcal{K} \downarrow d \to \mathcal{C}$ that maps $(c, f) \mapsto c$ with the obvious action on morphisms. This functor losses a lot of information about object but it picks those objects whose morphism preserve the structure of the comma category.

Up to this point we have uploaded a kind of hologram of d into $\mathcal{C}$. Now, we need to transport this information to $\mathcal{E}$ using $\mathcal{F}$. Thus, we have a map,

$$\mathcal{F} \circ \Pi_d : \mathcal{K} \downarrow d \to \mathcal{E}$$

This is a digram in $\mathcal{E}$ that somehow approximate the image of d. So, lastly we need to choose from $\mathcal{E}$ an object that glue or best represents all this information together. In principle, this is just a cone under the digram (cocone). So now we can give a formulae that defines the action of this functor,

$$Lan_{\mathcal{K}}\mathcal{F}(d) = colim(\mathcal{F} \circ \Pi_d : \mathcal{K} \downarrow d \to \mathcal{E})$$

However, to prove something like that ones need to check that this is indeed a functor, construct the associated natural transformation and prove the claimed universality property. One can check [12] theorem 6.2.1 for details. Thus, now one can reach the following corollary,

Corollary 2.1.1. If $\mathcal{K} : \mathcal{C} \to \mathcal{D}$ is a functor so that $\mathcal{C}$ is small and $\mathcal{D}$ is locally small then, If $\mathcal{E}$ is cocomplete, then the left kan extension exist and is given by the colimit formula.

2.5 Operads

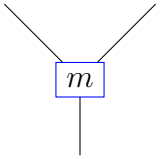
As per the motivation, we want to formalize the notion of this disk algebras. To make this work we need a convenient method to encode this algebraic structure that is independent of the type of object which we want to realize. This is the framework described by an operad. An operad encodes the information of the algebraic structure by specifying the algebraic operations and the rules one needs to follow.However, in our case this is not enough, since in particular we still have this notion of homotopy and operad relations are strict rules. Thus

we wish again to weaken this to make these relations hold up to homotopy this homotopy version of operads are called ∞-operads. One typically will observe new structures that arise during this passage. Between the classical associative and commutative operads there is an infinite tower of operads, known as the $\mathbb{E}_n$ algebras that converge in some sense to the commutative operad $\mathbb{E}_\infty$. [5]. For the simplicity of the discussion, we will give a notion of the operads of interest which are the framed disc operads and leave the homotopy version since it require much more analysis.

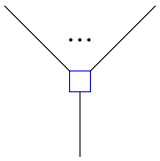
Loosely speaking operads keep track of various types of operations. Consider for example an algebra map,

$$m : V \times V \to V$$

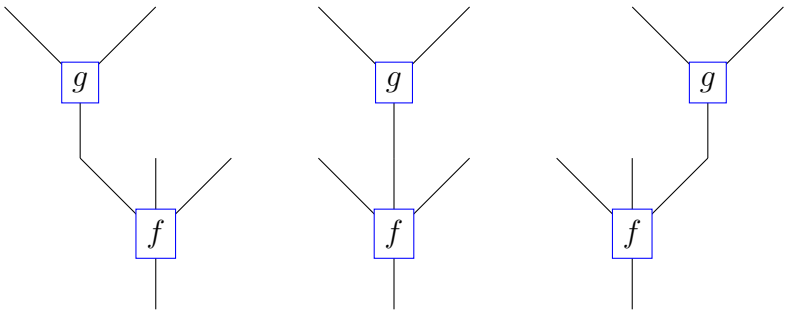
we can further ask m to be associative or commutative. Behind all these operations is a particular operad. One can think of m as a $2 \to 1$ map, graphically represented by the following tree,



Now, we want to generalize this notion by encoding operations $n \to 1$ which is called n-ary operation.



Now suppose that such operations do exist. We wish for a consistent way to compose these operations. Let us consider f to be a 3-ary and g a 2-ary. We can compose these operation by putting the output of g in any input of f which graphically look like,



Thus, in general there are n-ways to compose m-ary operation with n-ary operation to produce $m+n-1$-ary operation. An operad is just the collection of all such operation with some compatibility conditions which we can describe as follows,

Definition 2.13. An operad in a category $\mathcal{S}$ consists of a sequence of objects $\mathcal{P}(k)$, $k \in \mathbb{N}$ a unit map $\eta : 1 \to \mathcal{P}(k)$ a right action of the symmetric group Σ_k on $\mathcal{P}(k)$ for each k and product maps

$$\gamma : \mathcal{P}(k) \boxtimes \mathcal{P}(n_1) \boxtimes \cdots \boxtimes \mathcal{P}(n_k) \to \mathcal{P}(n_1 + \cdots + n_k)$$

which are associative, unital and Σ-equivalent. meaning that the following digrams commute,

1. Associativity

$$\begin{array}{ccc}
\mathcal{P}(k)\boxtimes(\boxtimes_{s=1}^{k}\mathcal{P}(n_s))\boxtimes(\boxtimes_{r=1}^{n}\mathcal{P}(j_r)) & \xrightarrow{\gamma\boxtimes id} & \mathcal{P}(n)\boxtimes(\boxtimes_{r=1}^{n}\mathcal{P}(j_r)) \\
\downarrow{\scriptstyle shuffle} & & \\
\mathcal{P}(k)\boxtimes(\boxtimes_{s=1}^{k}\mathcal{P}(n_s)\boxtimes\boxtimes_{t=1}^{i_s}\mathcal{P}(j_{i_{s-1+t}})) & & \downarrow{\scriptstyle\gamma} \\
\downarrow{\scriptstyle id\boxtimes_s\gamma} & & \\
\mathcal{P}(k)\boxtimes(\boxtimes_{s=1}^{k}\mathcal{P}(h_s)) & \xrightarrow[\gamma]{} & \mathcal{P}(j)
\end{array}$$

   where $n = n_1 + \cdots + n_k$, $j = j_1 + \cdots + j_n$, $h_s = j_{i_{i_{s-1}+1}} + \cdots + j_{i_s}$ with $i_s = j_1 + \cdots + j_s$

2. Unit

$$\begin{array}{ccc}
1\boxtimes\mathcal{P}(k) & \xrightarrow{\cong} & \mathcal{P}(k) \\
\downarrow{\scriptstyle\eta\boxtimes id} & \nearrow{\scriptstyle\gamma} & \\
\mathcal{P}(1)\boxtimes\mathcal{P}(k) & &
\end{array}
\qquad
\begin{array}{ccc}
\mathcal{P}(k)\boxtimes 1^k & \xrightarrow{\cong} & \mathcal{P}(k) \\
\downarrow{\scriptstyle id\boxtimes\eta^k} & \nearrow{\scriptstyle\gamma} & \\
\mathcal{P}(k)\boxtimes(\boxtimes\mathcal{P}(1)^k) & &
\end{array}$$

3. Σ-equivalence,

$$\begin{array}{ccc}
\mathcal{P}(k)\boxtimes\mathcal{P}(n_1)\boxtimes\cdots\boxtimes\mathcal{P}(n_k) & \xrightarrow{\sigma\boxtimes\sigma^{-1}} & \mathcal{P}(k)\boxtimes\mathcal{P}(n_{\sigma(1)})\boxtimes\cdots\boxtimes\mathcal{P}(n_{\sigma(k)}) \\
\downarrow{\scriptstyle\gamma} & & \downarrow{\scriptstyle\gamma} \\
\mathcal{P}(n) & \xrightarrow{\sigma(n_1,\cdots,n_k)} & \mathcal{P}(n) \\
\mathcal{P}(k)\boxtimes\mathcal{P}(n_1)\boxtimes\cdots\boxtimes\mathcal{P}(n_k) & \xrightarrow{id\boxtimes\tau_1\boxtimes\cdots\boxtimes\tau_k} & \mathcal{P}(k)\boxtimes\mathcal{P}(n_{\sigma(1)})\boxtimes\cdots\boxtimes\mathcal{P}(n_{\sigma(k)}) \\
\downarrow{\scriptstyle\gamma} & & \downarrow{\scriptstyle\gamma} \\
\mathcal{P}(n) & \xrightarrow{\tau_1\oplus\cdots\oplus\tau_k} & \mathcal{P}(n)
\end{array}$$

   where $\sigma \in \Sigma_k$, $\tau_i \in \Sigma_{n_i}$, $\sigma(n_1, \cdots, n_k) \in \Sigma_{n_1,\cdots,n_k}$ permutes k blocks as σ permutes k letter and $\tau_1 \oplus \cdots \oplus \tau_k \Sigma_{n_1+\cdots+n_k}$ is a block sum of permutations.

All of the axioms are clear from the topological side. There are two basic example of operads.

Example 2.6. Let V be a vector space. Define,

$$End_V(k) = Hom(V^{\boxtimes k}, V)$$

The space of multilinear maps $V^k \to V$. Then, $End_V(k)$ forms an operad called the endomorphism operad where the composition is given by the of outputs of a map g into the ith input of another map f and the action of the symmetric group is clear.

Thus an operad is a blueprint of operations where it's operations not just for a fixed arity (like binary) but for all kinds of operations—binary, ternary, and so on—and it comes equipped with rules that dictate how you can compose these operations. Operads exist for their algebras. An algebra over an operads is a realization of the rules into a mathematical object like a vector space.

Definition 2.14. Let $\mathcal{P}$ be an operad in a category $\mathcal{S}$. An algebra over an $\mathcal{P}$ is an object $X \in \mathcal{S}$ for which the operations encoded by the operad is realized. X is equipped with maps,

$$\theta_k : \mathcal{P}(k) \boxtimes X^{\boxtimes k} \to X$$

which are again associative unital and Σ-equivariant.

The other important example is the little n-disk operad,

Example 2.7. Let D^n be the n-dimensional open unit disc in $\mathbb{R}^n$. Define the spaces $\mathcal{D}_n(k)$ to be the embedding of k disjoint union of disks inside another disk,

$$f : \coprod_1^k D^n \to D^n$$

Here f is a composition of translations and dilations only. The operad maps is given by the composition,

$$\gamma(f, g_1, \cdots, g_k) = \coprod_{n_1+\cdots+n_k} D^n \xrightarrow{g_1 \amalg \cdots \amalg g_k} \coprod_k D^n \xrightarrow{f} D^n$$

Thus, one can think of $D(k)$ as a disc with k holes and the composition can be thought of as sticking the k-disks, g_i in the k holes of f.

An example for algebras over the little disk operad is the n-fold loop space,

$$\Omega^n Y = \{f : (D^n, \partial D^n) \to (Y, \star)\}$$

which is the space of all continuos maps from n-dimensional cube with certain boundary to topological space Y. Then the maps, $\theta_k : D_n(k) \times (\Omega^n Y)^k \to \Omega^n Y$ is obtained by evaluating the ith n-fold loop on the ith embedded disc (f_i will be evaluated on disc i in $D_n(k)$).

Example 2.8. Framed disc operads: In similar fashion to the definition of the little disc operad the framed disc operad $fD_n(k)$ consists of the embeddings from the disjoint union of k n-dimensional discs obtained by composing translations, dilations and rotations. one can think of the elements of $fD_n(k)$ as a configuration of the discs with marked points to distinguish rotations.

3 Factorization Homology

Now, that we have discussed the framework in which the desired functor can operate, we are ready to define factorization homology as in [2]. Factorization homology is essentially this extension functor that associates a manifold invariant by associating data to a disc system of local coordinates in an ∞-category and in some sense integrates this over the

manifold (Meaning we are extending the data of disk algebra functor to the whole manifold). Specifically, the factorization homology is the left kan extension along this disc algebra system.

The factorization homology takes as an input an n-dimensional manifold with an E_n algebra which is an algebra over the little disc operad over a symmetric monoidal ∞-category $\mathcal{C}$. Ayala and Francis proved that factorization homology satisfy Eilenberg-Steenrod axioms for singular homology [2]. Thus, one may formally consider factorization homology as a generalization of singular homology made for topological manifolds. In particular, the factorization homology satisfy excision. Furthermore, it was shown that depending on the algebraic input one can obtain different homology theories [3]. Notice now that in our category $\mathfrak{Mfld}_n$, the hom set of embedding of manifold $\mathcal{N}$ into $\mathcal{M}$ is a topological space (∞-groupoid). We will now give a brief discussion on the main theorems and definitions.

Definition 3.1. An E_n-algebra is a symmetric monoidal functor $\mathcal{F} : \mathfrak{Disk}_n^{or} \to \mathcal{C}$, where $\mathcal{C}$ is a symmetric monoidal ∞-category. We further define $\mathcal{A} := \mathcal{F}(\mathbb{R}^n)$.

Now for the left kan extension to exist one has to require some properties on the target category, specially existence of some colimits. One can ask for $\mathcal{C}$ to be cocomplete. However, Ayala and Francis have used some stronger conditions, specifically $\mathcal{C}$ has to be $\otimes$-presentable.

Definition 3.2. A symmetric monoidal ∞-category $\mathcal{C}$ is $\otimes$-presentable if

- $\mathcal{C}$ is presentable. $\mathcal{C}$ admits colimits and every object is a filtered colimit of compact objects.
- For each $C \in \mathcal{C}$, the functor $C \otimes - : \mathcal{C} \to \mathcal{C}$ carries colimits digrams to colimit digrams.

Definition 3.3. Let $\mathcal{M}$ be an orientable n-manifold. Let $\mathcal{A}$ be $\mathfrak{Disk}_n^{or}$-algebra in $\otimes$-presentable category $\mathcal{C}$. The factorization homology of $\mathcal{M}$ with coefficients in $\mathcal{A}$ is an object in $\mathcal{C}$ given by the left kan-extension as,

$$\int_{\mathcal{M}} \mathcal{A} = colim(\mathfrak{Disk}_{n/\mathcal{M}} \to \mathfrak{Disk}_n \to \mathcal{C})$$

where, $\mathfrak{Disk}_{n/\mathcal{M}}$ is essentially our usual comma category or the category over $\mathcal{M}$ considered as simplicial set.

3.1 Properties of factorization homology

As mentioned before factorization homology satisfies excision. That mean that over a manifold, we can split the manifold intro two parts and evaluate the functor on each part independently.

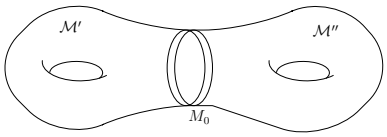
Definition 3.4. A collar gluing for an oriented n-manifold $\mathcal{M}$ is a continuos function,

$$f : \mathcal{M} \to [-1, 1]$$

such that the restriction on $(-1, 1)$ is a manifold bundle (meaning its fibred over every point in the interval and locally trivial). The collar gluing is often dented as an open cover,

$$\mathcal{M} = \mathcal{M}' \bigcup_{M_0 \times \mathbb{R}} \mathcal{M}''$$

where, $f^{-1}([-1, 0)) = \mathcal{M}'$, $f^{-1}((0, 1]) = \mathcal{M}''$ and $f^{-1}(0) = M_0$



Theorem 3.1. *Let $\mathcal{M} = M \times \mathbb{R}$ for some $(n-1)$ dimensional manifold M. Then, then the factorization homology is equipped with a monoidal structure making $\int_{M\times\mathbb{R}} \mathcal{A}$ an E_1-algebra.*

Proof. (sketch) The monoidal structure is clear. One just need to consider the embedding of

$$(M \times \mathbb{R}) \coprod (M \times \mathbb{R}) \to (M \times \mathbb{R})$$

which retract two copies of $(M\times\mathbb{R})$ into one. Thus, given M there is an induced symmetric monoidal functor,

$$\mathfrak{Disk}_{1,or} \to \mathfrak{Mfld}_{n,or}$$
$$\mathbb{R}^{\amalg_k} \mapsto M \times \mathbb{R}^{\amalg_k}$$

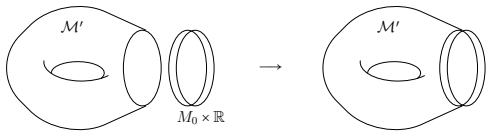
since the embedding $\mathbb{R}^{\amalg_k} \mapsto \mathbb{R}$ induce a map $M\times\mathbb{R}^{\amalg_k} \mapsto M\times\mathbb{R}$, this makes the factorization homology an E_1-algebra.

■

Theorem 3.2. *Let $\mathcal{M}$ have a collar gluing $\mathcal{M}' \bigcup_{M_0\times\mathbb{R}} \mathcal{M}''$ then, each factor $\int_{\mathcal{M}'} \mathcal{A}$ and $\int_{\mathcal{M}''} \mathcal{A}$ is a module over the algebra $\int_{M_0\times\mathbb{R}} \mathcal{A}$.*

Proof. Since $M_0 \times \mathbb{R}$ is a collar of $\mathcal{M}'$, one gets the embedding,

$$\rho : \mathcal{M}' \coprod (M \times \mathbb{R}) \to \mathcal{M}'$$



Now, by monoidality of $\int_{_} \mathcal{A}$, this translates into,

$$\int_{\mathcal{M}' \amalg (M_0\times\mathbb{R})} \mathcal{A} \cong \int_{\mathcal{M}'} \mathcal{A} \otimes \int_{M_0\times\mathbb{R}} \mathcal{A} \to \int_{\mathcal{M}'} \mathcal{A}$$

which gives $\int_{\mathcal{M}'} \mathcal{A}$ a right $\int_{M_0\times\mathbb{R}} \mathcal{A}$- module structure. In a similar fashion, $\int_{\mathcal{M}''} \mathcal{A}$ is a left module.

■

Remark. For the above to make sense, we need to define what is the tensor product over algebra in an ∞-category. This is the relative tensor product which is the colimit of the 2-sided bar construction.

Now, we state the excision property of factorization homology.

Theorem 3.3. *[2] Let $\mathcal{C}$ be a symmetric monoidal ∞-category which is $\otimes$-presentable. Then, factorization homology valued in $\mathcal{C}$ satisfies $\otimes$-excision: For any $\mathfrak{Disk}_n$-algebra $\mathcal{A}$ in $\mathcal{C}$ and for collar gluing $\mathcal{M} \approx \mathcal{M}' \bigcup_{M_0 \times \mathbb{R}} \mathcal{M}''$, there is an equivalence in $\mathcal{C}$,*

$$\int_{\mathcal{M}'} \mathcal{A} \bigotimes_{\int_{M_0 \times \mathbb{R}} \mathcal{A}} \int_{\mathcal{M}''} \mathcal{A} \to \int_{\mathcal{M}} \mathcal{A}$$

Furthermore, factorization homology is a pointed theory meaning it comes with a canonical E_0-structure. This can be constructed as follow; Let $\varnothing$ be the empty manifold, since $\varnothing$ is the monoidal unit in $\mathfrak{Mfld}_n$ and the factorization homology is a monoidal functor, then we have the isomorphism,

$$\int_{\varnothing} \mathcal{A} \approx 1_{\mathcal{C}}$$

But we note that $\varnothing$ is initial in $\mathfrak{Mfld}_n$. This directly translates into a unique morphism,

$$1_{\mathcal{C}} \to \int_{\mathcal{M}} \mathcal{A}$$

by the factorization homology.

3.2 Factorization homology of punctured surfaces valued in Rex

In this section, we will focus on the case of 2-dimensional manifolds (surfaces). Furthermore, we will briefly discuss main findings of [13]. First of all, we recall that the factorization homology associates the same object to homeomorphic manifolds. Thus, any manifold that is homeomorphic to a disk would immediately be mapped to same object given by the disc algebra in $\mathcal{C}$. Furthermore, by recalling that all 2-dimensional surfaces can be classified by their genus and boundary components, one can reach the following proposition,

Proposition 3.1. Let $\mathcal{A}$ be an E_2-algebra in $\mathcal{C}$. Then, the functor $\int_{_} \mathcal{A}$ is such that if U is contractable, then there is an equivalence in $\mathcal{C}$,

$$\int_{U} \mathcal{A} \approx \mathcal{A}$$

Next, we need to choose a target category $\mathcal{C}$ that is $\otimes$-presentable. For applications in representation theory, [13] choose the category **Rex**, the category of finitely co-complete $\mathbb{K}$-linear categories with right exact functors (functors that preserve finite colimits or equivalently for abelian categories functors that preserve direct sum and co-kernels thus taking an exact sequence $A \to B \to C \to 0$ to an exact sequence [14]). We refer to [15] for basis definition of the elements and morphism inside this category. **Rex** is $\otimes$-presentable. Thus, it can be used as our target category. Furthermore, **Rex** is (2,1)-category, thus all the machinery we have for ∞-categories just reduces down to the case of truncated (2,1)-category. **Rex** is also of particular interest since as we will show later we can explicitly compute the pointwise kan-extension for any punctured surface. Additionally, **Rex** is symmetric monoidal by the use of Deligne-Kelly tensor product.

Definition 3.5. Let $\mathcal{C}$, $\mathcal{E}$ and $\mathcal{D}$ be $\mathbb{K}$-linear categories and let $Bilin(\mathcal{C}\times\mathcal{D},\mathcal{E})$ denote the category of $\mathbb{K}$-bilinear functor from $\mathcal{C}\times\mathcal{D}$ to $\mathcal{E}$ that preserve finite colimits in each variable separately with natural isomorphisms as morphisms. Then, the Deligne-Kelly tensor product $\mathcal{C}\boxtimes\mathcal{D}$ of $\mathcal{C}$ and $\mathcal{D}\in\mathbf{Rex}$ is defined by the universal property,

$$\mathbf{Rex}[\mathcal{C}\boxtimes\mathcal{D},\mathcal{E}]\approx Bilin(\mathcal{C}\times\mathcal{D},\mathcal{E})$$

Now, let us intuitively describe what one expects to obtain when computing the factorization homology for a punctured surface Σ with a single boundary interval. Notably as illustrated in Fig.1, one have the embedding of disks inside Σ which identifies the boundary part with part of the boundary of the disk resulting in a surface homeomorphic to Σ.

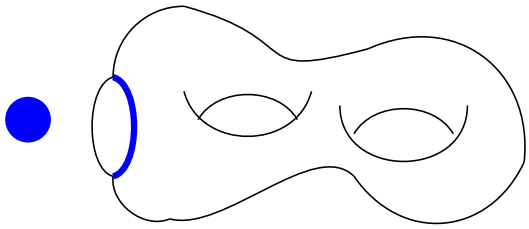


Figure 1: Figure illustrating the embedding of a disk on the boundary of a surface resulting in a surface which is homeomorphic to $\Sigma_{2,1}$

Now again by monoidality and functionality of the factorization homology, this gives $\int_\Sigma\mathcal{A}$ an $\mathcal{A}$-module structure (By definition we have $\int_{D^2}\mathcal{A}\approx\mathcal{A}$). Now, to describe the evaluation of the functor on the surface Σ, we need to have a description for the category $\mathcal{A}$ which essentially describes the E_2-algebra. Furthermore, we need to find a way to construct this module. But beforehand, we need to probably define modules over tensor categories. Module categories generalize the notion of modules over a ring.

Definition 3.6. [15] Let $\mathcal{A}$ be a tensor category in **Rex**, A right $\mathcal{A}$-module category $\mathcal{M}$ for a tensor category $\mathcal{A}$ is a locally finite category in **Rex** with an action functor in **Rex**,

$$\triangleleft:\mathcal{M}\boxtimes\mathcal{A}\to\mathcal{M}$$

which satisfies the standard pentagon and triangle axiom. We denote $\triangleleft(m\boxtimes a)$ by $m\otimes a$. Similar definition can be made for left $\mathcal{A}$-module category.

Definition 3.7. [15] Let $\mathcal{C}$ and $\mathcal{D}$ be tensor categories. A $(\mathcal{C},\mathcal{D})$-bimodule category is a category $\mathcal{M}$ that has left $\mathcal{C}$-module and right $\mathcal{D}$-module structures with module associativity constraints compatible by a collection of natural isomorphisms called the middle associativity constraints.

Definition 3.8. Let $\mathcal{M}$ and $\mathcal{N}$ be two module categories over $\mathcal{A}$. An $\mathcal{A}$-module functor from $\mathcal{M}$ to $\mathcal{N}$ consists of a functor $F:\mathcal{M}\to\mathcal{N}$ with a natural isomorphism,

$$s_{a,m}:F(m\otimes a)\to a\otimes F(m)$$

and satisfies the standard pentagon and triangle axioms.

Remark. To fully respect the abelian structure in **Rex**, the action functor has to be a module bifunctor that is bilinear on morphisms. Furthermore, it has to be right exact, colimit preserving, in both variables. One should also notice that **Rex** is equivalent as a category to $\mathbf{Pr}_c$ of compactly generated presentable categories with compact co-continuous functors. For am accessible discussion on $\mathbf{Pr}_c$ one can check [16].

Since we have this no-trivial equivalence, on recalls the following theorem.

Proposition 3.2. (**Adjoint functor theorem**) [17] Let $F : \mathcal{C} \to \mathcal{D}$ be a functor between locally presentable categories, then F has a right adjoint if and only if it preserves small colimits (right exact).

This means that for a tensor category $\mathcal{A}$ and for $\mathcal{M}$ a $\mathcal{A}$-module category in **Rex** the functor,

$$a \mapsto (a \otimes m) : \mathcal{A} \to \mathcal{M}$$

has a right adjoint, which we denote by Hom (as by conventional notation to right adjoint of tensor functors),

$$n \mapsto Hom(m, n) : \mathcal{M} \to \mathcal{A}$$

Thus, there exist an object $Hom(m_1, m_2) \in \mathcal{A}$ and natural isomorphism,

$$\mathbf{Hom}_{\mathcal{M}}(a \otimes m_1, m_2) \cong \mathbf{Hom}_{\mathcal{A}}(a, Hom(m_1, m_2))$$

Definition 3.9. The object $Hom(m_1, m_2)$ that represents the above functor is called the internal Hom from m_1 to m_2.

The action functor is by definition a module functor. However, this is not always the case for its right adjoint. Nonetheless, one can show that it is a module functor given that $\mathcal{A}$ is rigid.

Proposition 3.3. Given that $\mathcal{A}$ is rigid. The assignment $n \mapsto Hom(m, n) : \mathcal{M} \to \mathcal{A}$ is an $\mathcal{A}$-module functor.

Proof. We need to show that $Hom(m, a \otimes n) \cong a \otimes Hom(m, n)$. Consider the following series of isomorphisms,

$$\begin{aligned}
\mathbf{Hom}_{\mathcal{A}}(b, Hom(m, a \otimes n)) &\cong \mathbf{Hom}_{\mathcal{M}}(b \otimes m, a \otimes n) \\
&\cong \mathbf{Hom}_{\mathcal{M}}(a^* \otimes (b \otimes m), n) \\
&\cong \mathbf{Hom}_{\mathcal{M}}((a^* \otimes b) \otimes m, n) \\
&\cong \mathbf{Hom}_{\mathcal{A}}(a^* \otimes b, Hom(m, n)) \\
&\cong \mathbf{Hom}_{\mathcal{A}}(b, a \otimes (Hom(m, n)))
\end{aligned}$$

Then by Yoneda lemma we reach the equivalence.

■

Remark. One should notice that the right adjoint to the action does not yet belong to **Rex** since in particular we need to show that its right exact.

Nonetheless, let us consider this adjunction in $\mathbf{Cat}^{\times}$. Formally, such an adjunction gives rise to a monad. Let us denote the functor $-\otimes m$ and $Hom(m,-)$ by L and R respectively. Then, one can form a monad on $\mathcal{A}$ using the endofunctor $T = R \circ L$ with the following unit and counit relations,

$$\eta : id_{\mathcal{A}} \to T \qquad m : T \circ T \xrightarrow{id \boxtimes \epsilon \boxtimes id} T$$

In this manner, one can find an equivalence for the category $\mathcal{M}$ (corresponding to $\int_{\mathcal{M}} \mathcal{A}$). This can be done by introducing an equivalence with the Eilenberg–Moore category through the comparison functor.

Definition 3.10. The Eilenberg–Moore category is the category $T - mod_{\mathcal{A}}$ of $T - algebras$ in $\mathcal{A}$ whose objects are pairs (a, f) of $a \in \mathcal{A}$ and morphism $f : T(a) \to a$ and morphisms from (a_1, f_1) to (a_2, f_2) are $h : a_1 \to a_2$ making the obvious digram commute.

Thus, one obtains a comparison functor $\Phi : \mathcal{M} \to T - mod_{\mathcal{A}}$ which send n to $R(n) = Hom(m,n) \in \mathcal{A}$, which has a canonical T action,

$$T \circ R(n) = R \circ L \circ R(n) \xrightarrow{id \boxtimes \epsilon} R(n)$$

Now, one would naturally ask when this comparison functor is an equivalence or equivalently when is R monadic. This is the general result of Barr-Beck theorem.

Theorem 3.4. *[18] The functor Φ defined above is an equivalence if and only if,*

- *R is conservative/reflects isomorphisms.*
- *$\mathcal{M}$ has co-equalizers of R-split parallel pairs and R preserves those co-equalizers.*

Since, in particular, R-split co-equalizer are co-equalizers [19] and R is by definition right exact, so it preserves split co-equalizer, the second condition is trivially satisfied. However, we have still to check if R is conservative and right exact. For the former, one have the following result

Proposition 3.4. Let $\mathcal{D}$ be an abelian category and $Hom(m,-)$ is a right exact functor. Then $Hom(m,-)$ is conservative if and only if for any n with $Hom(m,n) \cong 0$, then $n \cong 0$.

Proof. Suppose $f : n \to l$ be a morphism such that $Hom(m,f)$ is an isomorphism. Since $Hom(m,-)$ is right exact, then it preserves finite colimits and since it is a right adjoint it also preserves arbitrary limits. Thus, one have $Hom(m, ker(f)) = kerHom(m,f) = 0$ and $Hom(m, coker(f)) = cokerHom(m,f) = 0$ which implies $ker(f) = coker(f) = 0$. Since, $\mathcal{D}$ is abelian this implies f is an isomorphism. The other direction follows in the same way. ∎

Thus, technically speaking if one requires such property on the right adjoint with right exactness, one would definitely reach the desired equivalence. However, stronger conditions were chosen in [13]. Technically, they require $Hom(m,-)$ to be faithful thus satisfying proposition 3.4 and colimit preserving thus trivially being right exact. This, in general, does totally depend on the choice of m. Thus, naturally one is lead to the following definition,

Definition 3.11. Let $\mathcal{A}$ be a $\mathbb{K}$-linear category in **Rex** with module category $\mathcal{M} \in \mathbf{Rex}$. We say $m \in \mathcal{M}$ is an $\mathcal{A}$-progenerator if the right adjoint $Hom(m, -)$ to the action functor is faithful and colimit preserving.

In general, there is no grantee that such m exists. However, if it does, one can formally construct $\mathcal{M}$

Proposition 3.5. Let $\mathcal{A}$ be a rigid abelian tensor category in **Rex**, and let $\mathcal{M} \in \mathbf{Rex}$ be an $\mathcal{A}$-module category with an $\mathcal{A}$-progenerator $m \in \mathcal{M}$. Then, the right adjoint functor is an equivalence of $\mathcal{A}$-module categories,

$$\mathcal{M} \cong End(m) - mod_{\mathcal{A}}$$

where $End(m) = Hom(m, 1_{\mathcal{A}} \otimes m)$

Proof. Indeed, following the assumption that m is an $\mathcal{A}$-progenerator then this, by proposition 3.4, makes $R = Hom(m, -)$ monadic. Then,

$$\mathcal{M} \cong T - mod_{\mathcal{A}}$$

where $T = Hom(m, - \otimes m)$. Now using proposition 3.3, this endofunctor is an $\mathcal{A}$-module functor. which implies $\forall a \in \mathcal{A}$, $T(a) = Hom(m, a \otimes m) = a \otimes Hom(m, 1 \otimes m) = a \otimes End(m)$. Thus, the monad is given by multiplication with the $End(m)$ object.

■

Although, we have our desired equivalence, we are in total doubt whether this progenerator exists or not. One may first ask if such progenerator does exist for the simplest A-module which is A itself. In that sense, we have the tensor functor $T^2 : \mathcal{A} \boxtimes \mathcal{A} \to \mathcal{A}$, $a \boxtimes a' \mapsto a \otimes a'$. It is, however, natural to define the action by tensoring with $1_{\mathcal{A}}$. Thus, one expects that $1_{\mathcal{A}}$ is the progenerator.

Proposition 3.6. The tensor unit $1_{\mathcal{A}}$ is a progenerator for the 2-fold right action of $\mathcal{A} \boxtimes \mathcal{A}$ on $\mathcal{A}$.

Proof. One have the action functor $\mathcal{A} \to \mathcal{A}$ ($a \mapsto a \otimes 1_{\mathcal{A}} \cong a$). Thus, we want to show that $Hom(1_{\mathcal{A}}, -)$ is faithful and colimit persevering. By invoking Yoneda lemma, we have $\mathbf{Hom}_{\mathcal{A}}(b, Hom(1_{\mathcal{A}}, a)) \cong \mathbf{Hom}_{\mathcal{A}}(b \otimes 1_{\mathcal{A}}, a) \cong \mathbf{Hom}_{\mathcal{A}}(b, a) \cong \mathbf{Hom}_{\mathcal{A}}(b, a \otimes 1_{\mathcal{A}})$ which show equivalence between,$- \otimes 1_{\mathcal{A}}$ and $Hom(1_{\mathcal{A}}, -)$. Thus, $Hom(1_{\mathcal{A}}, -)$ is faithful. To show the right adjoint is colimit preserving, we will show that T admits a right adjoint which is co-continuous in each variable. By the equivalence with $\mathbf{Pr}_c$ T is co-continuous functor. Thus, by proposition 3.2, T has a right adjoint T_R. Furthermore, T_R is cocontinuous if $\mathcal{A}$ is rigid [13].

■

Now, we know what is this progenerator for disks. However, this does not yet tell us whether we can find it for other surfaces. So we wish for a universal property to define this this progenerator. One then can notice that $1_{\mathcal{A}}$ is an image of the monoidal unit from any tensor category. We still want this arrow to be unique. Thus one has to choose $\mathbf{Vect}_k$ (The initial object in **Rex**) which would map $\mathbb{K} \mapsto 1_{\mathcal{A}}$. Now, using factorization homology this is just the embedding of $\varnothing \to \mathbb{R}^2$. This information indeed may seem trivial, however, as we will see it totally defines the factorization homology of the surface. So, one can define the following,

Definition 3.12. The image of the tensor unit $\mathbb{K} \in \mathbf{Vect}$ under the canonical functor induced by the embedding $\varnothing \to \mathcal{M}$ is called the distinguished object and denoted $\mathcal{O}_{\mathcal{A},\mathcal{M}} \in \int_{\mathcal{M}} \mathcal{A}$.

Even though, we have the above construction, these is no canonical way to determine the distinguished object before knowing the category itself. Thus, one has to rely again on some topological notions to construct this category. For surfaces, one recalls that any surface is equivalent to gluing of disks in some manner (attachment of 2-handles). Let us now define how one can formally make such surface,

Definition 3.13. A gluing pattern is a bijection,

$$P : 1, 1', \cdots, n, n' \to 1, \cdots, 2n$$

such that $P(i) < P(i')$

In that sense the gluing patten determines how the handles get attached to a disk to form the desired surface. Topologically, as illustrated in Fig.2, one can form any surface using some gluing pattern.

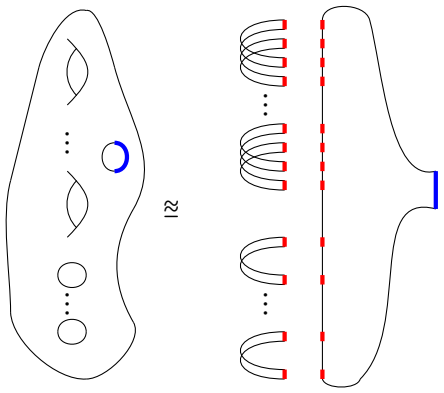


Figure 2: Gluing pattern of a surface with genus g and r boundary components

However, if one have this equivalence. Then using excision property, one can formally calculate the factorization homology of the surface by dividing the surface into three parts a disjoint union of handles, some twisting of the arms of the handles according to the gluing pattern and finally a marked disk.

However, excision needs the set of g-handles to have an $\mathcal{A}^{\boxtimes 2n}$-module structure. For example, if we have only two handles, then we need to have $(a_1 \boxtimes a_2) \boxtimes (b_1 \boxtimes b_1' \boxtimes b_2 \boxtimes b_2') \mapsto (a_1 \otimes b_1 \otimes b_1') \boxtimes (a_2 \otimes b_2 \otimes b_2')$. Thus, as in the case for algebras, one needs to endow $\mathcal{A}$ with a braiding structure to swap the two middle components. Furthermore, if we require $\mathcal{A}$ to be braided and rigid, this makes $\mathcal{A}$ an $\mathcal{A}^{\boxtimes k}$-module, where shuffle is just a composition of elements $\sigma_{a_1 \otimes \cdots \otimes a_k, b_i}$. Furthermore, for the $2n$-marked disc, this should also have an $\mathcal{A}^{\boxtimes 2n}$-module structure. As in proposition 3.6, one can show that $1_{\mathcal{A}}$ is the progenerator of the n-fold right/left action of $\mathcal{A}^{\boxtimes n}$ on $\mathcal{A}$.

Remark. We can swap between right and left actions on $\mathcal{A}$ since $\mathcal{A}$ is rigid and the category is braided.

Nonetheless, we still need to define the notion of relative tensor product over the tensor category $\mathcal{A}$. This is basically a generalization of the usual tensor product over algebras.

Definition 3.14. A bilinear functor $F : \mathcal{M} \times \mathcal{N} \to \mathcal{C}$ is $\mathcal{A}$-balanced if there is a natural isomorphism which on components is,

$$i_{m,a,n} : F(m \triangleleft a, n) \to F(m, a \triangleright n)$$

satisfying a standard pentagon axiom.

Definition 3.15. [20] Let $\mathcal{A}$ be a $\mathbb{K}$-linear tensor category, $\mathcal{M}$ and $\mathcal{N}$ be right and left $\mathcal{A}$-module categories respectively. The relative tensor product $\mathcal{M} \boxtimes_{\mathcal{A}} \mathcal{N}$ is a $\mathbb{K}$-linear category together with an $\mathcal{A}$-balanced functor out of $\mathcal{M} \times \mathcal{N}$ satisfying the universal property,

$$\mathbf{Rex}[\mathcal{M} \boxtimes_{\mathcal{A}} \mathcal{N}, \mathcal{E}] \approx Bal_{\mathcal{A}}(\mathcal{M} \times \mathcal{N}, \mathcal{E})$$

where $Bal_{\mathcal{A}}(\mathcal{M} \times \mathcal{N}, \mathcal{E})$ is the functor category of $\mathcal{A}$-balanced functor with $\mathcal{A}$-balanced natural transformations.

Before proceeding to check the equivalence we described above, let us prove the following lemma,

Lemma 3.5. *Let $\mathcal{A}$ be a rigid abelian tensor category. Let $\mathcal{M}$, and $\mathcal{N}$ be right and left $\mathcal{A}$-module categories respectively with m and n their $\mathcal{A}$-progenerators. Then, there is an equivalence of categories.*

$$\mathcal{M} \boxtimes_{\mathcal{A}} \mathcal{N} \cong End(m) - mod_{\mathcal{N}}$$

In particular if $F : \mathcal{A} \to \mathcal{B}$ is a functor with faithful right adjoint, then

$$\mathcal{M} \boxtimes_{\mathcal{A}} \mathcal{B} \cong F(End(m)) - mod_{\mathcal{B}}$$

Proof. First one can see that $\mathcal{M}$ is dualizable. As for every $m \in \mathcal{M}$, one can formally construct $Hom(m, -) \in \mathcal{M}^{\vee}$. Furthermore, one have an evaluation,

$$ev : \mathcal{M}^{\vee} \boxtimes \mathcal{M} \to \mathcal{A}$$

Since $\mathcal{M}$ is a right module, $\mathcal{M}^{\vee}$ is a left module by $(Hom(m, -) \cdot a)(n) = Hom(m, a \cdot n)$. Consider the following functor,

$$G : \mathcal{M} \boxtimes_{\mathcal{A}} \mathcal{N} \to Fun_{\mathcal{A}}(\mathcal{M}^{\vee}, \mathcal{N}) \to \mathcal{N}$$

$$m \boxtimes_{\mathcal{A}} n \mapsto Hom(m, -) \otimes n \mapsto Hom(m, m) \otimes n = End(m) \otimes n$$

The first map is an equivalence of categories similar to case of the modules over a ring [15]. The second arrow is conservative since m is an $\mathcal{A}$-progenerator. Then, G is conservative. Furthermore, G has a left adjoint given by $n \mapsto m \boxtimes_{\mathcal{A}} n$. Hence, we have the monad,

$$GG^{L}(n) : n \mapsto m \boxtimes_{\mathcal{A}} n \mapsto End(m) \otimes n$$

Now, we have G a right adjoint which is conservative in cocomplete category. Thus, by Barr-beck theorem, we get the equivalence,

$$\mathcal{M} \boxtimes_{\mathcal{A}} \mathcal{N} \cong End(m) - mod_{\mathcal{N}}$$

The category of $End(m)$-modules in $\mathcal{N}$. Now, if we have a functor $F : \mathcal{A} \to \mathcal{B}$, then $\mathcal{B}$ is a left $\mathcal{A}$-module category by the action, $(a, b) \mapsto F(a) \otimes b$. Thus, following the same proof as above with $\mathcal{N} = \mathcal{B}$ and,

$$G : m \boxtimes_{\mathcal{A}} b \mapsto F(Hom(m, -)) \otimes n \mapsto F(Hom(m, m)) \otimes n = F(End(m)) \otimes n$$

Since F has a faithful right adjoint, this ensures that the functor G is conservative, proposition 3.4. Thus, one gets,

$$\mathcal{M} \boxtimes_{\mathcal{A}} \mathcal{B} \cong F(End(m)) - mod_{\mathcal{B}}$$

■

Now, we are ready to check the equivalence,

Theorem 3.6. *Let $\mathcal{A}$ be an abelian rigid balanced braided tensor category in* **Rex**. *We have an equivalence of categories,*

$$\int_{\Sigma(P)} \mathcal{A} \cong A_{\Sigma(P)} - mod_{\mathcal{A}}$$

where $\Sigma(P)$ is the surface constructed using gluing pattern P and $A_{\Sigma(P)} = End(\mathcal{O}_{\mathcal{A},\Sigma(P)})$.

Proof. As illustrated in Fig.2, we can deform the punctured surface $\Sigma(P)$ into handles attached to left of a disk with $2n+1$ markings. Then, applying the factorization homology on the parts, we get,

$$\int_{D^2} \mathcal{A} \approx \mathcal{A}$$

and due to the markings on this disk, this induces a $\mathcal{A}^{\boxtimes 2n} - \mathcal{A}$-bimodule structure.

$$(a_1 \boxtimes \cdots \boxtimes a_{2n}) \boxtimes b \boxtimes a' \mapsto a_1 \otimes \cdots \otimes a_{2n} \otimes b \otimes a'$$

which we will denote, ${}_{2n}\mathcal{A}_1$. Similarly, for each handle H_i (homeomorphic to a disk), we have,

$$\int_{H_i} \mathcal{A} \approx \mathcal{A}_{\mathcal{A}^{\boxtimes 2n}}$$

That have a regular right action of $\mathcal{A} \boxtimes \mathcal{A}$-module structure, denoted $\mathcal{A}_2$. Thus, for the whole set of handles we get,

$$\int_{H_1 \cup \cdots \cup H_n} \mathcal{A} \cong \mathcal{A}^{\boxtimes n}$$

which is a $\mathcal{A}^{\boxtimes 2n}$-module category. However, due to the gluing pattern, it induces the following structure,

$$(a_1 \boxtimes \cdots a_n) \boxtimes (b_1 \boxtimes \cdots \boxtimes b_{2n}) \mapsto (a_1 \otimes b_{P(1)} \otimes b_{P(1')}) \boxtimes \cdots \boxtimes (a_n \otimes b_{P(n)} \otimes b_{P(n')})$$

which is denoted $\mathcal{A}^P$. Thus, by excision property, one gets,

$$\int_{\Sigma(P)} \mathcal{A} \cong A^P \underset{\mathcal{A}^{\boxtimes 2n}}{\boxtimes}{}_{2n}\mathcal{A}_1$$

Now, let us look closely at the category $\mathcal{A}^P$. This is the category of tensoring n copies of $\mathcal{A}$, then applying a permutation. However, each copy of these disks has $1_{\mathcal{A}\mathcal{A}}$ as its progenerator. Thus, before applying the permutation we have,

$$\mathcal{A}^{\boxtimes n} \cong End(1_{\mathcal{A}\mathcal{A}})^{\boxtimes n} - mod_{\mathcal{A}^{\boxtimes 2n}}$$

This is mainly because factorization homology is monoidal. Applying the permutation to this would just swap the factors in the multiplication of the monad. Thus let $End(1_{\mathcal{A}\mathcal{A}})^P$ denote multiplication element after applying the permutation on the monad. Thus, we can write,

$$\mathcal{A}^P \cong End(1_{\mathcal{A}\mathcal{A}})^P - mod_{\mathcal{A}^{\boxtimes 2n}}$$

However, using the tensor functor $T^{2n} : \mathcal{A}^{\boxtimes 2n} \to \mathcal{A}$, which has a faithful right adjoint, one can use lemma 3.5 and replace $\mathcal{B}$ with $\mathcal{A}$ and $\mathcal{M}$ with $\mathcal{A}^P$, thus we end up with,

$$A^P \underset{\mathcal{A}^{\boxtimes 2n}}{\boxtimes}{}_{2n}\mathcal{A}_1 \cong T^{2n}(End(1_{\mathcal{A}\mathcal{A}})^P) - mod_{\mathcal{A}}$$

as a right $\mathcal{A}$ module category. Now, we need to check that this element is indeed $End(\mathcal{O}_{\mathcal{A},\Sigma})$. Using the pointing structure of factorization homology, we have,

$$\mathbf{Vect} \to \mathbf{Vect} \boxtimes \mathbf{Vect} \to \mathcal{A}^{\boxtimes n} \boxtimes \mathcal{A} \to A^P \boxtimes \mathcal{A} \to \mathcal{A}^P \underset{\mathcal{A}^{\boxtimes 2n}}{\boxtimes} \mathcal{A}$$

which produces the distinguished element. The idea is to show that this element is our progenerator. Now, one can notice that $1_{\mathcal{A}^P}$ is the progenerator of $\mathcal{A}^P$ and the functor on factorization homology induced by the embedding of $\mathcal{A}^P$ is equivalent to the tensor functor. This is follows from the equivalence between locally constant factorization algebras on the disk and E_2-algebras. ([13], 5.2). Thus, using the above map, we have $1 \boxtimes 1$ gets mapped to an element in $\int_\Sigma \mathcal{A}$. However, since this map from **Vect** is unique, this has to be the distinguished element $\mathcal{O}_{\mathcal{A},\Sigma}$. Thus, this implies our equivalence.

■

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